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Densitaxis: Active particle motion in density gradients

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Organisms often swim through density-stratified fluids. Here, we investigate the dynamics of active particles swimming in fluid density gradients and report theoretical evidence of taxis as a result of these gradients (densitaxis). Specifically, we calculate the effect of density stratification on the dynamics of a force- and torque-free spherical squirmer and show that density gradients induce reorientation that tends to align swimming either parallel or normal to the gradient depending on the swimming gait. In particular, swimmers that propel by generating thrust in the front (pullers) rotate to swim parallel to gradients and hence display (positive or negative) densitaxis, while swimmers that propel by generating thrust in the back (pushers) rotate to swim normal to the gradients. This work could be useful to understand the motion of marine organisms in ocean or be leveraged to sort or organize a suspension of active particles by modulating density gradients.

microswimmers | taxis | active matter | fluid dynamics | biophysics

Significance

Bodies of water such as the ocean tend to feature gradients in density due to variations of temperature or salinity. Global warming is acting to increase this stratification. This work demonstrates how density gradients can lead to a form of taxis for microswimmers (densitaxis) and gives a simple mechanism to explain the resulting dynamics. These dynamics may affect the vertical migration of organisms in the ocean, as well as oceanic mixing, and suggest a mechanism that may be used to sort active particles.

Organisms often swim through inhomogeneous environments experiencing gradients in heat, light, nutrients, temperature, or salinity. They frequently respond to these inhomogeneities by reorienting and changing speed, exhibiting a directed motion called taxis. Active particle motion in gradients of light (1–3), nutrients (4–6), and recently viscosity (7–12), have been relatively well studied and even exploited to sort or organize both living and synthetic active matter systems (13–17). However, motion through density gradients and any resulting taxis is relatively less well understood despite the prevalence of density stratification in lakes, ponds, and oceans. Density gradients in aquatic bodies are generally caused by corresponding gradients in temperature or salinity and have been known to hinder the vertical motion of organisms like euphausiids and dinoflagellates (18, 19), play a role in the formation of algal blooms (20), and possibly reduce mixing induced by marine organisms (21, 22). Moreover, density stratification of the ocean has increased substantially in recent decades due to the effects of global warming (23).

The current understanding of (living or nonliving) swimmers in density gradients is largely based on the analyses of various model active particles in vertical gradients (see refs. 24–26 for a review). Initial studies found the flow due to a point-force or a force-dipole placed in a density-stratified fluid with negligible inertia and weak advection (27–29). Like in a homogeneous fluid, these singularity solutions represent the far-field flow due to a settling passive particle or a neutrally buoyant active particle in a stratified fluid. Analogous far-field flows with strong advection have also been recently found (22, 30).

Active particles in density gradients not only induce a distinct flow field but also swim at different speeds compared to homogeneous fluids. Vertically swimming particles have received much attention to address diel vertical migration of marine organisms in search of food. Studies have shown speed changes are indifferent to whether the particle swims up or down the density gradients, for instance, a swimming sheet that propels by passing traveling waves on its surface experiences a speed up or slowdown in density gradients relative to its speed in homogeneous fluids (31). On the other hand, a squirmer with negligible inertia slows down in density gradients if it generates thrust in the back (generally called pushers) but speeds up if it generates thrust in the front (pullers) (32, 33). These changes are caused by the buoyancy of the fluid displaced by the particle (22). Increasing fluid inertia tends to lead to a slowdown for both pushers and pullers (32, 34).

While past research has focused on active particles moving vertically in vertical density gradients, without knowledge of reorientation it is unknown whether active particles would tend to swim in a vertical orientation. Here, we address the question of whether microswimmers tend to align with density gradients from an arbitrary initial orientation as a result of hydrodynamics. Notably, we show that puller-type squirmers rotate to align

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vertically, displaying positive or negative densitaxis, but pushers rotate to swim horizontally. This implies that density gradients may aid vertical migration for pullers while hinder (or block) it for pushers. We also generalize earlier findings on velocity changes experienced by particles swimming vertically (33) to those experienced while swimming in arbitrary direction.

Results

We consider an active particle moving in an otherwise quiescent Newtonian fluid (see Fig. 1 for a schematic). The fluid density in the absence of the particle, $\rho_\infty(\mathbf{x})$, varies spatially due to a similar variation in the temperature or salt concentration. The background (or ambient) density usually varies over a length scale of kilometers in the ocean. At the scale of the small μm-mm sized particles we consider here, we take the ambient density to vary linearly,

$$\nabla\rho_\infty = \gamma\hat{\mathbf{g}}, \qquad [1]$$

where $\gamma > 0$ (for stable stratification) is the magnitude of the gradient and $\hat{\mathbf{g}} = \mathbf{g}/|\mathbf{g}|$ is a unit vector oriented in the direction of gravity. The quiescent ambient fluid, $\mathbf{u}_\infty = \mathbf{0}$, satisfies a hydrostatic equation $\nabla p_\infty = \rho_\infty \mathbf{g}$, where p_∞ is the ambient pressure field.

Generally one expects the density of organisms to be similar to their suspending fluid and for simplicity, we assume here that the particle density exactly matches the background fluid density, $\rho_p = \rho_\infty$ (32, 33, 35, 36). As the particle swims, it would hence need to alter its density according to the background density to stay density matched. Such density regulation can be achieved by employing a swim bladder or mechanisms like the carbohydrate ballasting (37) or the ion replacement (38, 39).

We model the active particle as a spherical squirmer of radius a and orientation $\mathbf{p}$. In the squirmer model, the shape of the swimmer is fixed and activity due to any swimming gait is represented simply by a prescribed tangential slip on the particle surface (40–42). This model is a reasonable representation of ciliated organisms like *Paramecium* and *Opalina* which propel by synchronously beating numerous cilia on their surface.

The flow in the presence of the particle is governed by the incompressible Navier–Stokes equations assuming the Boussinesq approximation [which holds for moderate density gradients (43–45)] where the density is taken as constant, $\bar{\rho}$, except in the weight of the fluid. $\bar{\rho}$ is taken as the mean density in a regime containing the particle that is large compared to the particle size. The flow field advects temperature or salinity fields whose transport is governed by an advection–diffusion equation that ultimately couples with the fluid density.

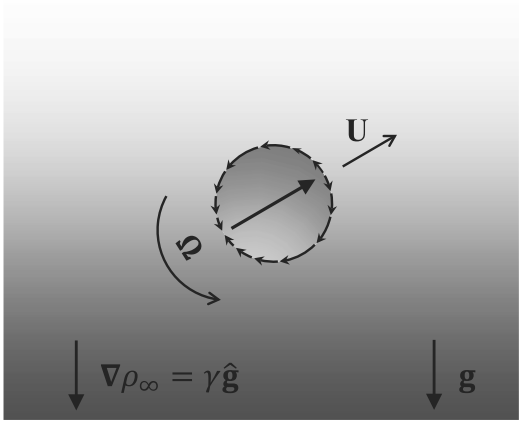

Fig. 1. An active particle in a fluid where the density varies linearly in the direction of gravity **g**.

The relevant characteristic scales of the problem are the particle size a, the density change across the particle γa, the swimming speed in a homogeneous fluid U_N, and the viscous pressure scale $\bar{\rho}\nu U_N/a$, where ν is the kinematic viscosity of the fluid. Using these, the governing equations are characterized by three dimensionless numbers: the Reynolds number, $Re = aU_N/\nu$, measures the relative importance of inertia to the viscous forces, the viscous Richardson number, $Ri_v = a^3N^2/\nu U_N$, measures the importance of buoyancy to the viscous forces, and the Péclet number, $Pe = aU_N/\kappa$, measures the importance of the advective to the diffusive transport rate of the density with diffusivity κ. The Prandtl number Pr is the ratio of the momentum to the mass diffusivity ν/κ and it can be written in terms of Re and Pe as $Pr = Pe/Re$. The Brunt-Väisälä (or buoyancy) frequency, $N = \sqrt{g\gamma/\bar{\rho}}$, appearing in Ri_v is the rate at which a vertically displaced inviscid fluid element oscillates in density gradients. For typical values of these dimensionless numbers, we consider planktonic organisms and ocean water. *Escherichia coli*, *Paramecium*, and *Daphnia* are of size 1 to 10 μm, 100 μm, and 5 mm, and they swim in water at speeds 10 μm/s, 1 mm/s, and 1 cm/s, respectively (46, 47). Hence the particle size a ranges from 1 μm to 5 mm while the particle speed U_N varies from 10 μm/s to 1 cm/s. On the other hand, the viscosity of water $\nu \approx 10^{-6}$ m^2/s while Pr ranges from 7 to 700 depending on whether variations in density are due to temperature or salinity, respectively. In oceans, lakes, and ponds, N varies from 10^{-4} to $0.3\,\mathrm{s}^{-1}$ (48, 49). These attributes can be used to estimate values of the Reynolds number, $Re = 10^{-5}$ to 50, the viscous Richardson number, $Ri_v = 10^{-15}$ to 1, and the Péclet number, $Pe = 10^{-4}$ to 10^4. Here, we focus on the regime where inertia is negligible, the limit $Re \to 0$, with weak but nonnegligible effects of buoyancy, and the advection of density. This parameter regime corresponds to the motion of small organisms (≈ 10 to 1,000 μm) in water stratified with weak gradients.

When advection is weak, density transport is dominated by diffusion near the active particle; however, no matter how small Pe there is always a distance, characterized by the screening length $l_\rho \sim a/Pe$, beyond which advection becomes on the order of diffusion, where $\sim$ denotes asymptotic equality. Likewise, no matter how small Ri_v, buoyancy forces become comparable to viscous forces at a distance from the particle characterized by the stratification screening length l_s (formally defined below). The flows generated by the active particle and its resulting dynamics depend on the relative order of magnitude of the two parameters, Pe and Ri_v (or l_s/l_ρ). We solve for the effect of stratification on the dynamics of the active particle only in the asymptotic regimes $l_s \ll l_\rho$, and $l_s \gg l_\rho$ in order to obtain analytical formulas. Details of our analytical methods are given in *Materials and Methods* below and expanded upon in *SI Appendix*.

In the regime $l_s \ll l_\rho$, far from the particle diffusion dominates advection. The stratification screening length is found by balancing buoyancy forces with viscous forces at $r \sim l_s$ to find $l_s \sim a\,(Ri_v Pe)^{-1/4}$. In this regime, we find the angular velocity induced by density stratification

$$\mathbf{\Omega} = c\beta\frac{U_N}{a}\,(Ri_v Pe)^{3/4}\,(\mathbf{p}\cdot\hat{\mathbf{g}})(\mathbf{p}\times\hat{\mathbf{g}}), \qquad [2]$$

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where β (which is dimensionless) indicates the swimmer type, $\beta > 0$ are pullers and $\beta < 0$ pushers, while the constant $c = E_E\left(1/\sqrt{2}\right)/4 - E_K\left(1/\sqrt{2}\right)/8 \approx 0.1059$ and E_K and E_E are the complete elliptic integrals of the first and second kind. Translational velocity changes are given in *SI Appendix*.

In the other regime, $l_s \gg l_\rho$. Far from the particle, now advection dominates diffusion. To determine the stratification screening length we again balance viscous forces and buoyancy forces to obtain $l_s \sim aRi_v^{-1/3}$. In this regime, we find the angular velocity induced by density stratification

$$\mathbf{\Omega} = c\beta\frac{U_N}{a}Ri_v\left(\mathbf{p}\cdot\hat{\mathbf{g}}\right)\left(\mathbf{p}\times\hat{\mathbf{g}}\right), \qquad [3]$$

where the constant of proportionality, $c \approx 0.09375$.

We see that in both asymptotic regimes, the angular velocity has the same functional form to leading order, writing rate of change of orientation $d\mathbf{p}/dt = \mathbf{\Omega}\times\mathbf{p}$ we have

$$\frac{d\mathbf{p}}{dt} = c\beta\varepsilon\frac{U_N}{a}(\mathbf{p}\cdot\hat{\mathbf{g}})(\mathbf{I}-\mathbf{pp})\cdot\hat{\mathbf{g}}, \qquad [4]$$

where only the constant of proportionality c (which is $\approx 10^{-1}$ in both cases) and ε (which is either $(Ri_vPe)^{3/4}$ or Ri_v) differ to leading order depending on the regime. In both cases, we see that fixed points in the dynamics are when $\mathbf{p} \parallel \hat{\mathbf{g}}$ and $\mathbf{p} \perp \hat{\mathbf{g}}$ with stability depending on the sign of β. In general, for pullers ($\beta > 0$) the stable orientation occurs when $\mathbf{p} \parallel \hat{\mathbf{g}}$, whereas for pushers ($\beta < 0$), it occurs when $\mathbf{p} \perp \hat{\mathbf{g}}$. In other words, in the presence of density gradients pullers will be reoriented to swim up or down the gradient (depending on whether the initial orientation is up or down), whereas pushers will be reoriented orthogonal to gradients. This is reminiscent of the far–field interaction of pushers and pullers with solid walls (50). In that case, the functional form is identical except the wall normal replaces the direction of gravity such that pushers orient parallel to the wall while pullers orient normal and the strength of the interaction decays with the distance from the wall, whereas in a constant density gradient, the far–field interaction with the stratified fluid is not a function of position.

Discussion

The mechanism for the density-induced rotation is conceptually similar to the far–field interaction of a swimmer with solid walls, and so despite the limits of the asymptotic calculation, we believe this mechanism is quite general. The rotation can be understood by analyzing the interaction of the far-field dipole flow and density isosurfaces (isopycnals), which we illustrate in a schematic in Fig. 2. Puller swimmers pull the fluid along their axis toward their body, ultimately perturbing the isopycnals from the horizontal in a manner shown in Fig. 2*A*. The tendency of isopycnals is to resist deformation and return to their stable state (gray arrows), this induces a flow (specifically baroclinic vorticity shown by blue curved arrows) that rotates the particle. Pushers do opposite to pullers, in that they push fluid along their axis and perturb isopycnals in an opposite manner, hence the induced baroclinic vorticity rotates them in the opposite direction (Fig. 2*B*). Swimmers also drive fluid normal to their axis which perturbs the isopycnals differently, but this perturbation and the ensuing baroclinic vorticity is weak and hence not shown here for clarity. Because the reorientation is ultimately determined by the far-field flow generated by the swimmer, it is insensitive to specific assumptions about the near-field details of the swimmer.

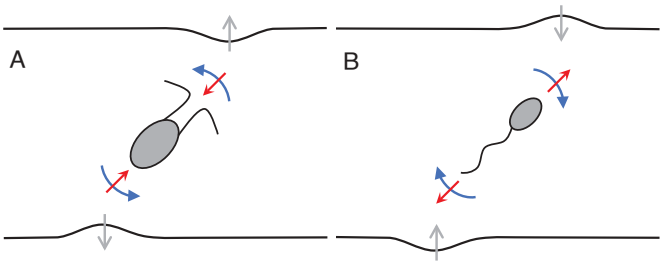

Fig. 2. Active particles deform isopycnals (black lines) differently for (*A*) pullers and (*B*) pushers. The resulting flow (baroclinic vorticity) induces reorientation.

This rotation occurs over a time scale $\tau = a/U_N c|\beta|\varepsilon$. If we assume the particles swim at roughly a body length per second, and $\beta = O(1)$ (12), we obtain $\tau \approx 10\,\mathrm{s}/\varepsilon$ depending on the value of ε (though we have assumed $\varepsilon^{1/3} \ll 1$). In a given density gradient (fixed γ or N), larger particles give larger ε and hence rotate faster. This densitactic rotation is unaffected by thermal fluctuations provided it occurs much faster than rotational diffusion $\tau_R^{-1} \equiv D_R = k_BT/8\pi\bar{\rho}\nu a^3$, i.e., $\tau \ll \tau_R$, where k_B is the Boltzmann constant and T is the absolute temperature. Using properties of water, $\tau < \tau_R$ leads roughly to $a > \varepsilon^{-1/3}\,\mu\mathrm{m}$. Taking $\varepsilon^{1/3} = 0.1$ as the largest value permissible for our asymptotic approach to be valid, we obtain $a > 10\,\mu\mathrm{m}$. Hence, the influence of fluctuations decreases with increasing particle size and matters for particles smaller than $O(10\,\mu\mathrm{m})$ in density gradients relevant to oceanic settings $N = 10^{-4}$ to $0.3\,\mathrm{s}^{-1}$. Factors other than thermal fluctuations may also contribute to the effective rotational diffusivity of the swimmer, such as biological noise, hydrodynamic interactions, or collisions with other organisms. In general, one can define a Langevin parameter $\chi_R = \tau_R/\tau$ that characterizes the balance between the strength of densitaxis and the effects that randomize the particle orientation, and even in the regime where $\chi_R < 1$ we expect there will still be a systematic bias in the dynamics due to the persistent densitactic reorientation (16, 51, 52).

In a similar manner, comparing densitaxis with rotation due to background shear $\dot{\gamma}$, for $\tau < \dot{\gamma}^{-1}$ we find $\dot{\gamma} < 10^{-4}\,\mathrm{s}^{-1}$. However, in establishing these bounds, we have been rather conservative with ε (to ensure the accuracy of our leading order perturbation formulas), but this is not a *physical* restriction, and if we allow ourselves to push the parameter limits of our asymptotics, we obtain the times scales as small as $\tau \approx 40\,\mathrm{s}$ for *Paramecium*, $\tau \approx 90\,\mathrm{s}$ for *Volvox* and $\tau \approx 50\,\mathrm{s}$ for the dinoflagellate *Ceratium furca* (see *SI Appendix* for further details), which leads to a relevant shear rate threshold $\dot{\gamma} < 0.01\,\mathrm{s}^{-1}$, which lies well within the shear rates found in nature (53, 54). Time scales much larger than these values are also possible for weak density gradients.

We note that the results obtained here assume that the particle density matches the surrounding background fluid. However, even if the particle is initially density matched, at some point it may *taxi* far enough to have a small density offset with the surrounding fluid. In that case, the far-field flow would be altered by the presence of a net force on the particle. We have found that to leading order the disturbance due to a net force does not contribute to the reorientation due to symmetry. Higher-order terms do contribute, and the degree of impact depends on

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the asymptotic regime and the magnitude of density mismatch, but preliminary work indicates that the effect is weak unless the density mismatch is significant, and then acts to reorient the particle toward neutral buoyancy (see *SI Appendix* for details). We also assume the particle is not bottom-heavy. Such particles display gravitaxis even in homogeneous fluids by rotating to swim vertically upward. Naively, we can compare time scales for gravitaxis reported in homogeneous fluids with the densitaxis found here (55, 56). Swimmers like *Chlamydomonas reinhardtii*, and *Paramecium* have been reported to have gravitactic time scales $\tau_g = 28\,\mathrm{s}$, and $\tau_g = 4$ to $14\,\mathrm{s}$, respectively. One might expect gravitaxis to dominate for *Chlamydomonas reinhardtii* given its small size, but for organisms like *Paramecium*, which can reach speeds of ten body lengths per second and values of ε at the large end of our regime of validity, the order of magnitude of the time scale for densitaxis can approach that of gravitaxis. The interplay of these two taxis could possibly enable swimmers to adopt an equilibrium orientation that is intermediate to vertical and horizontal. But this interplay is more subtle than simply comparing the rotation rate of each mechanism as a bottom-heavy torque would alter the far-field representation of a swimmer, and hence, the densitaxis of such swimmers could be different from that found here. We have also neglected the effects of variations in viscosity on the dynamics of the swimmer, despite the fact that the relative change in viscosity due to temperature or salinity can be much larger than the corresponding relative changes in density. The reason is that the changes in dynamics due to viscosity stratification, or *viscotaxis*, are determined by differences in the viscosity at the scale of the particle $\sim O(a/L)$, where L is the length scale of the gradient (57), and this ratio is negligibly small when considering microorganisms and gradients in the ocean. Conversely, the fundamental length scale for the effects of density stratification l_s is typically orders of magnitude smaller than the length scale of the gradient (28, 32).

The densitaxis shown here could have several implications. It could aid or hinder the diel vertical migration (DVM) of planktonic organisms in ocean depending on their gait and orientation. Many organisms, from zooplankton and phytoplankton to large whales and sharks, perform diel vertical migration (58–61), wherein organisms typically swim up toward the surface of ocean at dusk to feed and then swim down toward the deeper ocean at dawn, although migration in the opposite direction also occurs. But despite the prevalence and importance of DVM, factors affecting this motion are not completely understood. Temperature and salinity gradients have been shown to influence DVM. Observations in both natural settings and laboratory experiments show that strong temperature and salinity gradients sometimes aid, but often hinder (or block) DVM (18, 19, 62, 63). Temperature or salinity may affect organisms directly, but also indirectly by altering the mechanical properties of the fluid (namely the density and viscosity). Our work shows that swimming dynamics are necessarily altered by spatial gradients in the density and this may play a role in the diel vertical migration of organisms, either enhancing or hindering it in strong temperature or salinity gradients, and with global warming increasing ocean stratification, we believe it is important to study this mechanism further.

Densitaxis could also affect the distribution of a dilute suspension of active particles in density gradients, and hence, the mixing induced by them. Pullers tend to a nematic state pointing upward and downward with equal probability, while pushers tend to swim in horizontal planes (with isotropic in-plane orientation). These distributions may be important to incorporate into previous models of mixing by microswimmers that assumed an isotropic distribution (29). A further research avenue to investigate would be densitaxis for larger aquatic organisms with finite inertia. Inertia could induce additional rotation that competes with densitaxis to yield nontrivial equilibrium swimmer orientations (36). Finally, densitaxis might also be leveraged as a mechanism to separate a mixture of pushers and pullers by imposing density gradients by spatially varying salinity or temperature.

Materials and Methods

The introduction of even a passive particle into density-stratified fluid will generally disturb the background density, as the properties of the particle, such as thermal conductivity or salt diffusivity, usually differ from those of the fluid. An active particle will moreover generate a flow due to activity that then transports the density. The density field in the presence of the particle is therefore written $\rho(\mathbf{x}) = \rho_\infty(\mathbf{x}) + \rho'(\mathbf{x})$, where ρ' is the disturbance density. Likewise, we may write the pressure field in the presence of the particle as $p = p_\infty + p'$ and because we have assumed an otherwise quiescent fluid, $\mathbf{u} = \mathbf{u}'$.

The flow induced by the particle is governed by the incompressible Navier–Stokes equations assuming the Boussinesq approximation, where the density is taken as a constant $\bar{\rho}$, except in the weight of the fluid

$$\nabla \cdot \mathbf{u}' = 0, \qquad [5]$$

$$\bar{\rho}\left(\frac{\partial \mathbf{u}'}{\partial t} + \mathbf{u}' \cdot \nabla \mathbf{u}'\right) = \rho' \mathbf{g} - \nabla p' + \bar{\rho}\nu\nabla^2 \mathbf{u}'. \qquad [6]$$

$\bar{\rho}$ is the mean density of a region of fluid containing the particle that is significantly larger than the particle size.

Temperature or salinity transport is governed by an advection–diffusion equation, and when temperature or salinity variations are weak, gradients in temperature or salinity can be considered linearly proportional to gradients in density and so, in this limit, density transport is also governed by an advection–diffusion equation

$$\frac{\partial \rho'}{\partial t} + \mathbf{u}' \cdot \nabla \rho' + \gamma \mathbf{u}' \cdot \hat{\mathbf{g}} = \kappa \nabla^2 \rho'. \qquad [7]$$

As boundary conditions, we expect that the disturbance due to the particle vanishes far from the particle so that $\mathbf{u}' \to \mathbf{0}$, and $\rho' \to 0$ as $r = |\mathbf{r}| \to \infty$, where $\mathbf{r} = \mathbf{x} - \mathbf{X}$ with the center of the particle denoted $\mathbf{X}$. The fluid velocity at the particle surface (denoted $\partial\mathcal{B}$) satisfies the no-slip condition so that

$$\mathbf{u}'(\mathbf{x} \in \partial\mathcal{B}) = \mathbf{U} + \Omega \times \mathbf{r} + \mathbf{u}^s, \qquad [8]$$

where $\mathbf{U}$ and Ω are the unknown (and to be determined by dynamic conditions) rigid-body translational and rotational velocity of the particle while $\mathbf{u}^s$ is the velocity on the surface due to activity (deformation or slip).

The slip velocity is typically written as an expansion in Legendre polynomials

$$\mathbf{u}^s = \sum_n \frac{2}{n(n+1)} B_n P_n'(\mathbf{p} \cdot \mathbf{n})\mathbf{p} \cdot (\mathbf{I} - \mathbf{nn}), \qquad [9]$$

where $\mathbf{p}$ is the particle orientation, $\mathbf{n}$ is the outward unit normal to the particle surface, P_n is the Legendre polynomial of degree n, and $P_n'(x) = \frac{d}{dx}P_n(x)$. The polar slip coefficients, B_n, are often called "squirming" modes, while the azimuthal slip (or "swirl") is not considered here. In homogeneous Newtonian fluids, only the B_1 mode determines the swim speed $U_N = \frac{2}{3}B_1$ (we assume $B_1 \geq 0$ with no loss of generality) whereas the B_2 mode gives the slowest decaying flow and hence determines the far-field representation of the particle. The ratio of the first two modes is denoted by $\beta = B_2/B_1$, with $\beta > 0$ denoting pullers and $\beta < 0$ pushers. As shown later, particle rotation and taxis in density gradients depend only on the far-field representation of the particle, which

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to leading order is given by the B_2 mode. Hence, we keep only the first two squirming modes to analyze the effects of density gradients on swimming. This is also in line with the common practice in other analysis including swimming under confinement (64, 65), in complex fluids (66–71), and at finite inertia (72–75).

The boundary condition for the density disturbance at the particle surface depends on the scalar responsible for the density variations. We assume the scalar flux into the particle vanishes (impermeable to the salt or insulating to the temperature) which leads to a corresponding no-flux condition for density at the surface of the particle, $\mathbf{n} \cdot \nabla \rho = 0$ for $\mathbf{x} \in \partial\mathcal{B}$.

These equations are closed by conservation of linear and angular momentum of the particle, which determine the unknown rigid-body translational and rotational velocity, $\mathbf{U}$ and Ω, respectively.

We assume that inertia is negligible, the limit $Re \rightarrow 0$. The diffusivity and the viscosity of the fluid are assumed to be constant to leading order, even though they too can vary spatially due to the variations in temperature or salinity. Furthermore, given that we have assumed the particle density matches that of the fluid, the active particle has no net weight in the fluid and the particle inertia also vanishes, hence the disturbance hydrodynamic force and torque on the particle are zero,

$$\mathbf{F}' = \int_{\partial\mathcal{B}} \mathbf{n} \cdot \boldsymbol{\sigma}' dS = \mathbf{0}, \quad [10]$$

$$\mathbf{L}' = \int_{\partial\mathcal{B}} \mathbf{r} \times \left(\mathbf{n} \cdot \boldsymbol{\sigma}'\right) dS = \mathbf{0}, \quad [11]$$

where $\boldsymbol{\sigma}' = -p'\mathbf{I} + \bar{\rho}\nu \left\{\nabla \mathbf{u}' + \left(\nabla \mathbf{u}'\right)^T\right\}$. We also assume that the disturbance density field is steady in a frame of reference moving with the particle

$$\frac{D\rho'}{Dt} = \left(\mathbf{u}' - \mathbf{U}\right) \cdot \nabla \rho'. \quad [12]$$

With these assumptions our governing equations for the fluid become in *dimensionless* form. Equations below are also written in dimensionless form

$$\nabla \cdot \mathbf{u}' = 0, \quad [13]$$

$$-\nabla p' + \nabla^2 \mathbf{u}' + Ri_v\, \rho' \hat{\mathbf{g}} = \mathbf{0}, \quad [14]$$

$$\nabla^2 \rho' = Pe\left((\mathbf{u}' - \mathbf{U}) \cdot \nabla \rho' + \mathbf{u}' \cdot \hat{\mathbf{g}}\right). \quad [15]$$

We use a generalized reciprocal theorem to obtain explicit expressions for the particle velocities $\mathbf{U}$ and Ω. First, we note that the disturbance flow is governed by forced Stokes equations where the forcing is the weight of the fluid relative to background density. The dynamics of a force and torque-free active particle immersed in such a Stokes flow is given directly by the formula

$$\boldsymbol{U} = \bar{\boldsymbol{R}}_{\boldsymbol{FU}}^{-1} \cdot \left[\boldsymbol{F}_s + \boldsymbol{F}_f\right], \quad [16]$$

where we have used six-dimensional vectors for translational and rotational velocity, $\boldsymbol{U} = [\mathbf{U},\ \Omega]$ and force and torque, $\boldsymbol{F} = [\mathbf{F},\ \mathbf{L}]$. $\bar{\boldsymbol{R}}_{\boldsymbol{FU}}$ is the (6×6) resistance tensor for the particle in a homogeneous fluid of viscosity $\bar{\rho}\nu$, $\boldsymbol{F}_s$ is the thrust generated by an active particle in the same homogeneous fluid and $\boldsymbol{F}_f$ accounts for the changes due to the presence of density perturbations. The forces are obtained, using the reciprocal theorem (76, 77), by projecting onto a known rigid-body motion auxiliary flow (see *SI Appendix* for details).

Separating translational and rotational velocities we obtain

$$\mathbf{U} - \mathbf{U}_N = \frac{Ri_v}{8\pi} \hat{\mathbf{g}} \cdot \int_{\partial\mathcal{B}} \left[(\mathbf{I} + \mathbf{nn}) \int_1^\infty \rho' r dr + (\mathbf{I}/3 - \mathbf{nn}) \int_1^\infty \frac{\rho'}{r} dr\right] dS, \quad [17]$$

$$\Omega = \frac{Ri_v}{8\pi} \hat{\mathbf{g}} \times \int_{\partial\mathcal{B}} \mathbf{n} \int_1^\infty \rho' dr dS \quad [18]$$

where we have used the fact that when $Ri_v = 0$ the particle moves through a homogeneous Newtonian fluid where the dynamics of a squirmer are well known (78), $\mathbf{U} = \mathbf{U}_N = \mathbf{p}$ and $\Omega = \mathbf{0}$. We see that to evaluate the effects of density gradients, we simply need to integrate the disturbance density against a known kernel. It may seem as if we have bypassed the need to solve for the disturbance flow; however the disturbance flow couples to the disturbance density by way of Eq. **15**.

Scaling arguments can be used to determine that the dominant effect on particle rotation comes from the region far from the particle (see *SI Appendix* for details). As such, we derive equations valid in the far-field ($r \gg 1$), where appropriate singular forcing terms are added to Eqs. **13**–**15** to account for the presence of the force-free active particle to leading order in the far field. Specifically, a force dipole of strength $\mathbf{S} = 2\pi\beta\,(3\mathbf{pp} - \mathbf{I})$ and a degenerate quadrupole of strength $\mathbf{Q} = -2\pi\mathbf{p}$, for the disturbance flow, while the density disturbance generated by the particle has character of concentration-dipole $\mathbf{D} = -2\pi\hat{\mathbf{g}}$, coefficients are assumed constant to leading order. We linearize by discarding the nonlinear advection of the disturbance density by the disturbance flow (as the product of small terms), and approximate the particle velocity with its value in homogeneous fluid (which holds for weak density gradients $Ri_v \ll 1$) leading to

$$\nabla \cdot \mathbf{u}' = 0, \quad [19]$$

$$-\nabla p' + \nabla^2 \mathbf{u}' + Ri_v \rho' \hat{\mathbf{g}} = -\mathbf{S} \cdot \nabla \delta(\mathbf{r}) - \mathbf{Q} \nabla^2 \delta(\mathbf{r}), \quad [20]$$

$$\nabla^2 \rho' + Pe \mathbf{U}_N \cdot \nabla \rho' - Pe \mathbf{u}' \cdot \hat{\mathbf{g}} = -\mathbf{D} \cdot \nabla \delta(\mathbf{r}). \quad [21]$$

We solve these singularly forced linearized governing equations in Fourier space to find the Fourier transform of density for arbitrary values of Pe and Ri_v; however, we calculate the inverse transform, and hence velocities, only in the asymptotic regimes $l_s \ll l_\rho$, and $l_s \gg l_\rho$. Specifically, in the regime $l_s \ll l_\rho$ (where diffusion dominates advection far from the particle), we expand the density in Fourier space assuming $(Ri_v Pe)^{1/4} \ll 1$ ($l_s \gg 1$), and $Pe^{3/4}/Ri_v^{1/4} \ll 1$ ($l_s/l_\rho \ll 1$), then calculate the angular velocity Eq. **2** in real space using the convolution theorem. In the other regime, $l_s \gg l_\rho$ (where advection dominates diffusion far from the particle), we solve for the leading order density for $Ri_v^{1/3} \ll 1$ ($l_s \gg 1$) and $Ri_v^{1/3}/Pe \ll 1$ ($l_s/l_\rho \gg 1$) in Fourier space then numerically inverse transform to obtain the angular velocity Ω in real space and find that data fits Eq. **3**.

Expanded details of these methods may be found in *SI Appendix*.

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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